
Mean or Mode: What the Loss Chooses When Colour Is Ambiguous

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Abstract

Colourising a greyscale photograph has no single correct answer: a grey car is consistent with red, blue or white, and under a squared error the optimal prediction is then the mean of the plausible colours, which is close to grey. Two colourisers are trained on CIFAR-10 with everything but the loss held fixed, the second reading its predicted distribution at a temperature that interpolates between that distribution’s mean and its mode. Sweeping it traces a frontier on which accuracy and colour variety cannot both be maximised. At temperature one the classification network reproduces the squared error network pixel by pixel, 3.11 [3.07, 3.15] apart against the 12.49 [12.47, 12.52] separating either from the truth, and does so across independent seeds: the desaturation is a property of the question the loss asks, not of a model or a run.

1. Introduction

Given the lightness of a photograph, what colour was it? For many pixels the question has no answer: a greyscale car constrains its colour hardly at all, a greyscale frog constrains it a great deal. What exists is not a function from lightness to colour but a conditional distribution $p(a, b \mid L)$, wide in the first case and narrow in the second.

A network emits one colour per pixel, so something must collapse that distribution, and the loss decides what. Let $\mu = \mathbb{E}[Y \mid X = x]$. Adding and subtracting μ inside the squared error and expanding gives

$$\mathbb{E}[\|\hat{y} - Y\|^2] = \|\hat{y} - \mu\|^2 + \text{Var}(Y \mid X = x), \quad (1)$$

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since the cross term contains $\mathbb{E}[\mu - Y] = 0$. The second term is independent of the prediction, so the minimiser is $\hat{y} = \mu$ (Bishop, 2006): the model produces not a plausible colour but the average of the plausible ones, and the average of red, blue and white is a barely tinted grey.

2. Related work

Posing colourisation as classification over quantised ab bins, and decoding with an annealed mean, is due to Zhang et al. (Zhang et al., 2016); Larsson et al. (Larsson et al., 2016) independently predicted per-pixel colour histograms. That a squared error averages ambiguous targets is likewise established (Mathieu et al., 2016). Neither is claimed as new. What is added is a controlled measurement: everything but the loss is held fixed, the frontier is traced by one knob rather than two tuned systems, and the networks are compared against each other pixel by pixel. Three departures from (Zhang et al., 2016) follow: their class rebalancing is a second mechanism for raising vividness and would confound the trade measured here, so the cross-entropy is left plain; their fixed 313-bin grid becomes the same 10-unit grid cut to the cells the data visits; and CIFAR-10 replaces ImageNet.

3. Method

Work is in CIE Lab, which separates the lightness the network is given from the chroma it must guess and in which euclidean distance approximates perceived colour difference. The network receives L and predicts (a, b) everywhere.

Architecture. Six blocks of 3×3 convolution, batch normalisation (Ioffe & Szegedy, 2015) and ReLU at 64 channels, dilations 1, 1, 2, 4, 8, 1 (Yu & Koltun, 2016), then a 1×1 convolution to the output. Dilation rather than striding keeps full resolution and gives a receptive field of 35 pixels, wider than the image, so residual uncertainty is ambiguity in the data and not myopia. Everything up to the final convolution is shared; the first model minimises the squared error on (a, b) , the

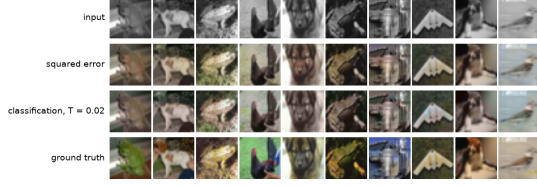


Figure 1. The two losses on the same test images, the classification model read at its mode. Both are muted against the truth, and muted in nearly the same way.

second quantises the *ab* plane into 10-unit cells, keeps the 124 the training set uses, and minimises the cross-entropy over them.

Readout. With bin centres c_k and probabilities p_k ,

$$\hat{y}(T) = \sum_k \frac{\exp(\log p_k/T)}{\sum_j \exp(\log p_j/T)} c_k, \quad (2)$$

the distribution’s mean at $T = 1$, which is what Equation (1) predicts the squared error model produces, and its mode as $T \rightarrow 0$.

Metrics and protocol. Colour error is the mean euclidean distance to the truth in *ab*; vividness is mean chroma against the truth’s own. Vividness alone is insufficient, since a model painting every pixel the same beige scores 39.0%, so variety, the mean distance from the model’s own average output and zero for any constant prediction, sits beside it. 3 seeds, 20 epochs, batch 128, AdamW (Loshchilov & Hutter, 2019) with a one-cycle schedule (Smith & Topin, 2019), in PyTorch (Paszke et al., 2019) on a laptop GPU. All numbers are on the 10 000 held-out test images of CIFAR-10 (Krizhevsky, 2009); labels are never shown to the colourisers. Code: <https://github.com/solazzo1911897-collab/colorization-mean-vs-mode>.

4. Results

Baselines and the cost of the mean. Grey everywhere gives 14.71 error; the average training colour gives 14.64 at 39.0% vividness and 0.0% variety, the pair that motivates reporting both. The squared error model reaches 12.44 [12.42, 12.47] at 62.0 [61.1, 63.1]% vividness and 53.8 [53.2, 54.4]% variety, beating both (Figure 1).

The frontier. Sweeping T (Figure 2) turns one prediction into a curve, and the squared error model sits on it: at $T = 1$ the classification model gives 12.43 [12.41, 12.46] and 53.9 [53.5, 54.3]% variety

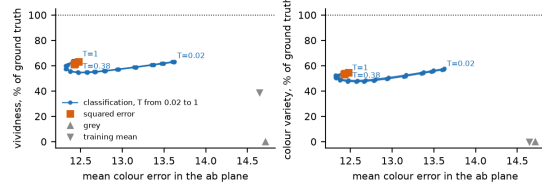


Figure 2. Accuracy against vividness (left) and variety (right) as the readout temperature is swept; all 3 seeds drawn. The constant baseline sits at 39.0% vividness and zero variety.

against 12.44 [12.42, 12.47] and 53.8 [53.2, 54.4]%. Toward the mode variety rises and accuracy falls, from 50.5 [50.2, 50.9]% at 12.32 [12.32, 12.33] ($T = 0.65$, lowest error) to 57.1 [56.7, 57.7]% at 13.61 [13.60, 13.63]. Neither end wins on both axes, and the squared error model cannot move along the curve at all. The range is narrow, though: even at the mode only 62.9 [62.8, 63.0]% of the truth’s vividness is kept against 62.0 [61.1, 63.1]%. With this much data the conditional distribution is concentrated enough that its mean and its mode nearly coincide, which is why the readout alone cannot close the gap. Vividness is not monotone in T , rising where the softmax flattens toward the bin grid’s centroid while variety does not.

The two models are one model. Equation (1) predicts the squared error output should coincide with the expectation of the classification model’s distribution, and it does: 3.11 [3.07, 3.15] in *ab* units between them against 12.49 [12.47, 12.52] between either and the truth, a factor of 4.0; against that model’s mode it is 6.78 [6.64, 6.96]. The control is the seed: pairs from different initialisations give 3.11 [3.08, 3.14], indistinguishable from the matched ones (Figure 3).

A prediction that failed. The desaturation was expected to be worst where colour is least predictable, yet grouped by CIFAR-10 class the correlation with vividness is +0.36 between instances and +0.02 over all pixels: no relation in the predicted direction. A per-pixel version using predictive entropy was also inconclusive (appendix).

Discussion. That the two models agree is what Equation (1) predicts, so the result is a confirmation rather than a surprise. What it adds is that the agreement survives finite data and runs sharing nothing but the loss, locating the desaturation in the objective rather than in any model. Vividness therefore cannot be bought by changing the loss: it has to come from the readout or from reweighting rare colours.

Statement on the use of AI

Anthropic’s Claude was used as an assistive tool throughout this project. It supported the refinement of the experimental methodology, assisted in extending and debugging the codebase, and aided in source verification and bibliographic assembly. It was also employed to polish, format, and synthesize drafts into the final report.

The Lab conversion was validated against published sRGB primaries, all citations were cross-checked at source, and all numerical values were derived directly from run logs. Failed predictions (e.g., Section 4) have been retained as observed.

References

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Appendix

The main limitations of this study are the resolution, at which colour ambiguity carries no texture component; a single dataset family; the absence of any perceptual evaluation, which for colourisation is what ultimately matters and requires human judgement rather than a distance in Lab; and class rebalancing, excluded by design, whose contribution to vividness is the obvious next measurement.

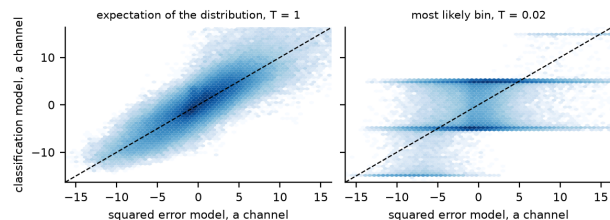


Figure 3. Squared error output against the classification readout, per pixel, on the a channel. Left, the expectation of its distribution, which agrees along the diagonal; right, its most likely bin, where the correspondence is gone and the discrete bin centres band the plot.

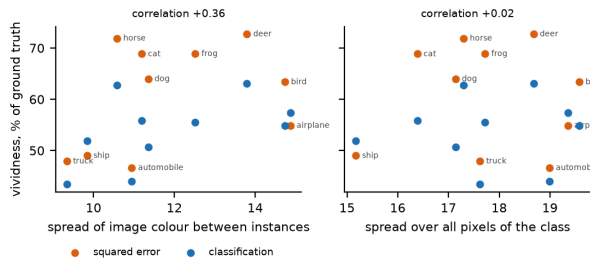


Figure 4. Vividness against two measures of how ambiguous the colour of a CIFAR-10 class is. Neither shows the predicted relation. Vividness is reported as a ratio to each class’s own ground truth, so classes dominated by large saturated regions of sky or sea carry a high denominator whatever their ambiguity, and the measure conflates the two. Kept because the prediction was made before the measurement.

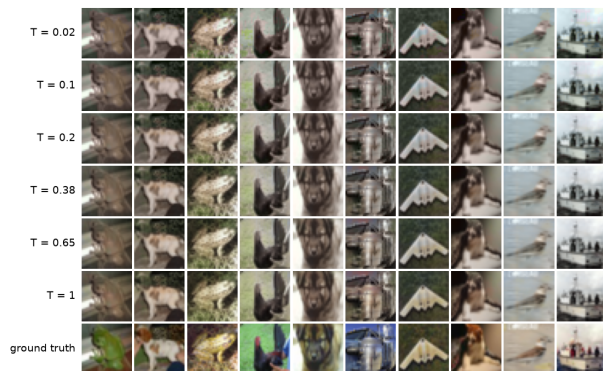


Figure 5. The same images decoded at increasing temperature, from the most likely colour to the mean of the distribution.